Applied Linear Algebra

Norm and Distance

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August 2026

Euclidean Norm

The *Euclidean norm* of an n-vector $\mathbf{x}$, denoted $\|\mathbf{x}\|$, is the squareroot of the sum of the squares of its elements,

$$\begin{aligned} \|\mathbf{x}\| &= \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \\ &= \sqrt{\mathbf{x}^\mathsf{T}\mathbf{x}} \\ &= \|\mathbf{x}\|_2 \end{aligned}$$

This is also called as the *magnitude* of a vector.

We will avoid calling it the *length* of a vector.

Euclidean Norm

$$\|\mathbf{0}\| = 0$$

$$\|\mathbf{e_i}\| = 1$$

$$\|[c]\| = c$$

$$\|\mathbf{1_n}\| = \sqrt{n}$$

we use terms such as a *small* norm and *large* norm when we want to compare the magnitudes of vectors.

Properties of Euclidean Norm

- Nonnegative homogeneity. $\|\beta\mathbf{x}\| = |\beta|\|\mathbf{x}\|$
- Nonnegativity. $\|\mathbf{x}\| \geq 0$
- Definiteness. $\|\mathbf{x}\| = \mathbf{0}$ only if $\mathbf{x} = \mathbf{0}$.
- Triangle inequality. $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

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Quiz:

Example for triangle inequality?

Norm - Wikipedia

a norm is a function from a real or complex vector space to the non-negative real numbers that behaves in certain ways like the distance from the origin: it commutes with scaling, obeys a form of the triangle inequality, and is zero only at the origin.

1-*norm* and ∞-*norm*

Definition
General Norm A *general norm* is any real-valued function of an n-vector that satisfies the four properties listed above.

Two examples of general norms:

$$\begin{aligned} \textbf{1-norm} \quad & \|x\|_1 &= |x_1| + \cdots + |x_n| \\ \textbf{Infinity-norm} \quad & \|x\|_\infty &= \max\{|x_1|, \dots, |x_n| \end{aligned}$$

Infinity norm $\|\cdot\|_\infty$ is exrensively used in ML-DSA, NIST's PQC signature scheme.

Root Mean Square Value

The norm is related to the root-mean-square (RMS) value of an n-vector x:

$$\mathbf{rms}(\mathbf{x}) = \sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}}$$

$$= \frac{\|\mathbf{x}\|}{\sqrt{n}}$$

The RMS value of a vector $\mathbf{x}$ is useful when comparing norms of vectors with different dimensions.

If all the entries of a vector are close to α, then the RMS value of the vector is $|\alpha|$.

Root Mean Square Value

The RMS value tells us what a 'typical' value of $|x_i|$ is.

The norm of $\mathbf{1}_n$, the n-vector of all ones, is $\sqrt{n}$, but its RMS value is 1, *independent* of n.

A couple of *norm identities*

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\| &= \sqrt{\|\mathbf{x}\|^2 + 2\mathbf{x}^\top\mathbf{y} + \|\mathbf{y}\|^2} \\ (\mathbf{x}+\mathbf{y})^\top(\mathbf{x}-\mathbf{y}) &= \|\mathbf{x}\|^2 - \|\mathbf{y}\|^2 \end{aligned}$$

Root Mean Square Value

- 1805 — Adrien-Marie Legendre. His criterion was to make the sum of the squares of the errors as small as possible.
- Karl Pearson is generally credited with introducing the term “root-mean-square” in 1895, although the underlying mathematical quantity, and even root-mean-square deviation, predates the terminology.
- Pearson’s important contribution was to name and formalize terminology around it.
- In his 1893 lectures, he introduced *standard deviation* as a replacement for the older *root mean square* error.

Euclidean Distance

If $\mathbf{a}$ and $\mathbf{b}$ are n-vectors,

$$\mathbf{dist}(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$$

The RMS value of the vector difference, *RMS deviation* between vectors $\mathbf{a}$ and $\mathbf{b}$ is

$$\begin{aligned} \text{RMSD} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(a_i - b_i)^2} \\ &= \frac{\mathbf{dist}(\mathbf{a}, \mathbf{b})}{\sqrt{n}} \end{aligned}$$

Euclidean Distance - Example

Given three vectors

$$\begin{aligned}\mathbf{u} &= [1.5, -2.5, 1.5, -1.5]\\ \mathbf{v} &= [0.5, -1.5, 3.5, -0.5]\\ \mathbf{w} &= [-1.5, 1.5, 2.5, -2.5]\end{aligned}$$

And the formula

$$\mathbf{dist}(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + (x_4 - y_4)^2}$$

$$\begin{aligned}\mathbf{u} - \mathbf{v} &= [1.0, -1.0, -2.0, -1.0]\\ \mathbf{dist}(\mathbf{u}, \mathbf{v}) &= \sqrt{(1.0)^2 + (-1.0)^2 + (-2.0)^2 + (-1.0)^2}\\ &= \sqrt{1.0 + 1.0 + 4.0 + 1.0}\\ &= \sqrt{7} \approx 2.65\end{aligned}$$

Euclidean Distance - Example

$$
\begin{aligned}
\mathbf{v} - \mathbf{w} &= [2.0, -3.0, 1.0, 2.0] \\
\mathbf{dist}(\mathbf{v}, \mathbf{w}) &= \sqrt{(2.0)^2 + (-3.0)^2 + (1.0)^2 + (2.0)^2} \\
&= \sqrt{4.0 + 9.0 + 1.0 + 4.0} \\
&= \sqrt{18} \approx 4.24
\end{aligned}
$$

$$
\begin{aligned}
\mathbf{u} - \mathbf{w} &= [3.0, -4.0, -1.0, 1.0] \\
\mathbf{dist}(\mathbf{u}, \mathbf{w}) &= \sqrt{(3.0)^2 + (-4.0)^2 + (-1.0)^2 + (1.0)^2} \\
&= \sqrt{9.0 + 16.0 + 1.0 + 1.0} \\
&= \sqrt{27} \approx 5.20
\end{aligned}
$$

Least Squares

For most choices of $\mathbf{b}$, however, there is no n-vector $\mathbf{x}$ for which $\mathbf{A}\mathbf{x} = \mathbf{b}$. As a compromise, we seek an x for which $\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$, which we call the *residual* (for the equations $\mathbf{A}\mathbf{x} - \mathbf{b}$), is as small as possible.

We should choose $\mathbf{x}$ so as to minimize the norm of the residual, $\mathbf{A}\mathbf{x} - \mathbf{b}$.

Least Squares

Minimizing the norm of the residual and its square are the same, so we can just as well minimize

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \|r\|^2 = r_1^2 + \cdots + r_m^2,$$

the sum of squares of the residuals.